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Plant Disease Segmentation Networks for Fast Automatic Severity Estimation Under Natural Field Scenarios

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Abstract: The segmentation of plant disease images enables researchers to quantify the proportion of disease spots on leaves, known as disease severity. Current deep learning methods predominantly focus on single diseases, simple lesions, or laboratory-controlled environments. In this study, we established and publicly released image datasets of field scenarios for three diseases: soybean bacterial blight (SBB), wheat stripe rust (WSR), and cedar apple rust (CAR). We developed Plant Disease Segmentation Networks (PDSNets) based on LinkNet with ResNet-18 as the encoder, including three versions: $\times$1.0, $\times$0.75, and $\times$0.5. The $\times$1.0 version incorporates a 4 $\times$ 4 embedding layer to enhance prediction speed, while versions $\times$0.75 and $\times$0.5 are lightweight variants with reduced channel numbers within the same architecture. Their parameter counts are 11.53 M, 6.50 M, and 2.90 M, respectively. PDSNetx0.5 achieved an overall F1 score of 91.96%, an Intersection over Union (IoU) of 85.85% for segmentation, and a coefficient of determination (R^2) of 0.908 for severity estimation. On a local central processing unit (CPU), PDSNetx0.5 demonstrated a prediction speed of 34.18 images (640 $\times$ 640 pixels) per second, which is 2.66 times faster than LinkNet. Our work provides an efficient and automated approach for assessing plant disease severity in field scenarios.

Keywords: deep learning; lightweight; plant disease; crop disease; semantic segmentation; severity estimation

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Academic Editor: Dimitre Dimitrov

Received: 15 February 2025
Revised: 2 March 2025
Accepted: 7 March 2025
Published: 10 March 2025

Citation: Zhao, C.; Li, C.; Wang, X.; Wu, X.; Du, Y.; Chai, H.; Cai, T.; Xiang, H.; Jiao, Y. Plant Disease Segmentation Networks for Fast Automatic Severity Estimation Under Natural Field Scenarios. *Agriculture* **2025**, *15*, 583. https://doi.org/10.3390/agriculture15060583

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1. Introduction

Plant diseases represent a substantial threat to global agricultural productivity, leading to estimated annual crop yield losses of 20–40%, which severely compromise food security and economic stability [1]. Among these, foliar diseases, the most prevalent type, significantly reduce yields by impairing photosynthetic efficiency. Traditionally, researchers have relied on manual visual estimation to assess disease severity in the field. However, this method is not only labor-intensive and time-consuming but also prone to human error [2,3]. Disease severity is quantified as the proportion of leaf area affected by disease spots. While image segmentation techniques, such as threshold segmentation, have been effective in isolating disease spots from healthy leaf tissue [4–6], they struggle with complex backgrounds and variable lighting conditions. Moreover, these methods necessitate the destructive collection of diseased leaves for laboratory imaging, which risks altering the disease's natural characteristics and is highly impractical. Additionally, detached leaves

often rapidly dehydrate and deform, complicating the application of these techniques. In contrast, deep learning approaches, particularly Convolutional Neural Networks (CNNs), offer an end-to-end solution. Once trained, CNNs can directly predict outcomes from input images without manual intervention.

With the rapid advancement of deep learning technologies, numerous classical network models [7–9] have been extensively employed in plant disease image recognition tasks. For instance, Long et al. [10] employed Convolutional Neural Networks to classify six types of wheat diseases in field environments, achieving an accuracy rate of 97.05%. Similarly, Bever et al. [11] applied the DenseNet model to classify eight types of soybean diseases, attaining a classification accuracy of 96.8%. Furthermore, Lin et al. [12] introduced a dual-branch network architecture that integrates attention mechanisms for simultaneous prediction, achieving an accuracy of 88.74% when classifying 10 distinct diseases.

In the field of disease severity assessment, several researchers have explored classification-based approaches to evaluate disease severity. For instance, Mi et al. [13] manually categorized wheat stripe rust images into six severity levels and employed a DenseNet model integrated with attention mechanisms, achieving a classification accuracy of 97.99%. Similarly, Joshi et al. [14] classified Vigna mungo disease images into three levels—healthy, moderate, and severe—using their custom VirLeafNet, attaining an accuracy of 97.4%. In a comparable study, Yin et al. [15] manually divided field maize disease images into five severity levels, achieving a classification accuracy of 97.12%. Despite the high classification accuracy of these methods, their dependence on manual severity level categorization introduces significant subjectivity, thereby limiting their reliability and broader applicability.

Another group of researchers has focused on the semantic segmentation of disease images with the goal of extracting lesion and leaf pixels from the images. For instance, Lin et al. [16] employed an enhanced U-Net [17] model to segment cucumber powdery mildew images, achieving an Intersection over Union (IoU) of 72.11%. Similarly, Li et al. [18] enhanced DeepLabv3+ [19] by integrating attention mechanisms and transfer learning, attaining an average IoU of 81.23% for cucumber disease image segmentation, and further assessed disease severity, achieving a coefficient of determination (R^2) of 0.958. Furthermore, Wang et al. [20] introduced a two-stage segmentation approach: the first stage utilizes DeepLabv3+ to segment diseased leaves in complex backgrounds, while the second stage applies U-Net for lesion segmentation. Despite the notable success of these methods in segmentation tasks, their models are characterized by substantial parameter sizes and high computational demands.

Model lightweight design has emerged as a prominent research focus in the field of deep learning. For instance, Feng et al. [21] introduced a U-shaped DFFANet model integrated with attention mechanisms for segmenting rice blast images, achieving an Intersection over Union (IoU) of 96.15% with a remarkably low parameter count of only 1.4 M. Similarly, Shi et al. [22] developed a U-Net model utilizing MobileNetV3 [23] as the encoder, incorporating attention mechanisms to segment wheat scab, achieving an IoU of 88.87% with a parameter count of 4.52 M. Furthermore, Xu et al. [24] implemented the DeepLabv3+network, leveraging channel scaling and knowledge distillation techniques, to perform two-stage segmentation of cotton diseases. This approach achieved IoU values of 87.66% and 85.18% for the two stages, respectively, with a parameter count of 5.02 M. Despite the notable success of these methods in minimizing network parameters and computational demands, their validation is limited to single-disease scenarios, and the lack of publicly available data poses significant challenges for reproducibility and broader application.

To address the limitations identified in current research, this study introduces improvements across three key areas: First, a comprehensive multi-disease dataset encompassing field scenarios was developed and publicly released. This dataset features high-resolution images of soybean bacterial blight (SBB), wheat stripe rust (WSR), and cedar apple rust (CAR). Second, leveraging the LinkNet network architecture, the PDSNets network family was proposed, which employs channel scaling techniques to achieve lightweight model design. Lastly, a quantitative evaluation of disease severity was performed based on the segmentation outcomes.

2. Materials and Methods

2.1. Data Acquisition and Preprocessing

This study utilized three types of disease images (Table 1), including SBB and CAR obtained from public resources, as well as WSR collected in the field by our team at the Jiaozuo Academy of Agriculture and Forestry Sciences (Jiaozuo, China) in May 2023. As illustrated in Figure 1, each image was padded from a rectangular format to a square format using black pixels and subsequently downsampled to 640 × 640 pixels using Lanczos interpolation [25]. A total of 80 images for each disease were processed using a custom Python 3.12.7 program. The integrated dataset is well suited for field plant disease segmentation due to its inclusion of the following characteristics: (1) high quality and resolution; (2) complex backgrounds and lighting conditions; (3) variable camera angles; (4) diseased leaves attached to the plant; and (5) diverse shooting equipment and locations. The dataset is publicly accessible at https://doi.org/10.34740/kaggle/ds/6620728, accessed on 9 March 2025.

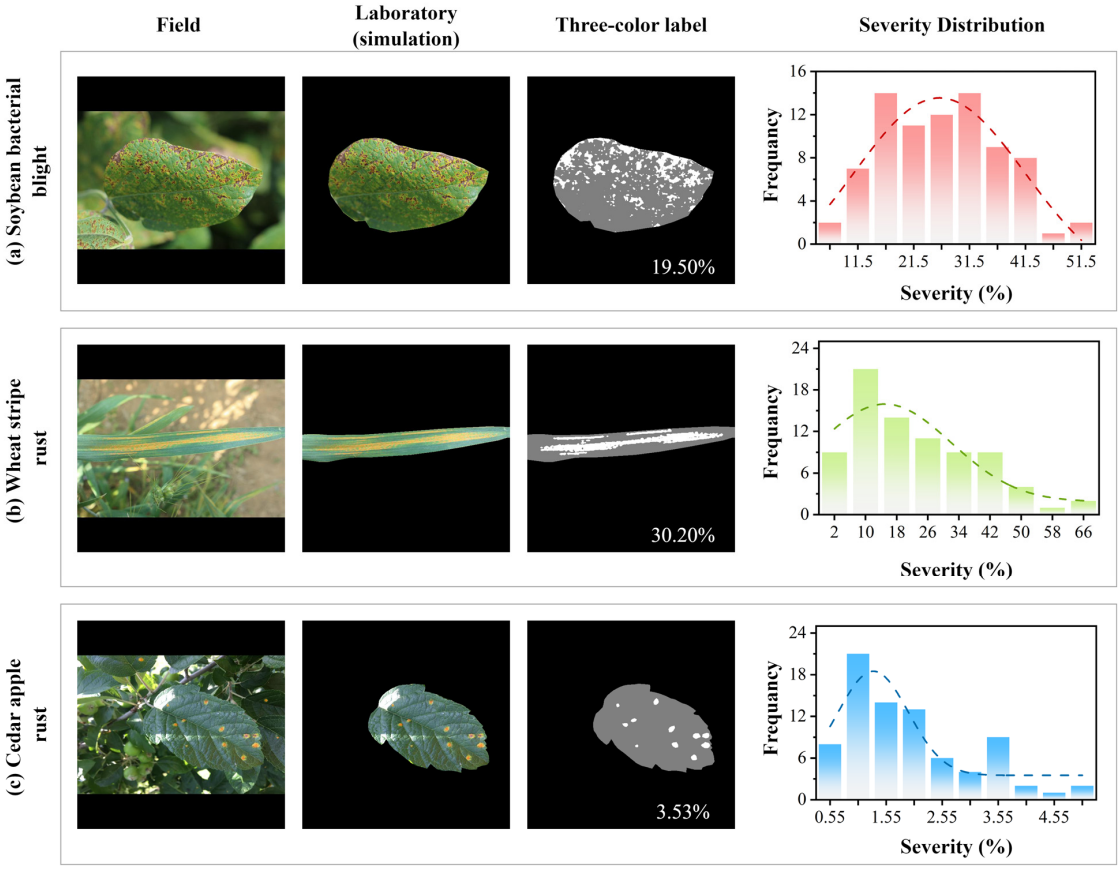


Figure 1. Three-color labels and reference severity distribution for disease images. Black, gray, and white in labels represent background (B), healthy leaf (H), and disease spots (S), respectively. Severity is calculated according to white × 100/(white + gray).

Table 1. Basic information of the original disease images.

Disease	Pixels	Year	Equipment	Location	Reference
Soybean bacterial blight	5472 × 3648	2020, 2021	Canon EOS 7D Mark II	Tallassee, AL, USA	[11]
Wheat stripe rust	4272 × 2848	2023	Canon EOS 450D	Jiaozuo, China	Our study
Cedar apple rust	2048 × 1365	2019	Canon Rebel T5i DSLR	Geneva, NY, USA	[26]

2.2. Pixel-Level Annotation

Before conducting supervised learning, each pixel of each image must be labeled as one of the pixel classes: background (*B*), healthy leaf (*H*), and disease spots (*S*). Severity is the proportion of a leaf occupied by disease spots.

$$Severity = \frac{S}{S+H} \times 100 \quad (1)$$

Due to the complexity of the data, we have different labeling schemes for each type of disease, as shown below. These schemes were inspired by reference [4] and were adapted during the annotation process. The implementation of all schemes was based on self-developed Python 3.12.7 programs.

SBB images contain leaves with regular margins and spots with an extremely complex distribution (Figure 1a). We used LabelMe v5.7.0 software [27] to manually label the polygonal boundary of a major leaf, thus setting the B pixels to black. Then, we converted the black background image from RGB color space to LAB color space and took its a-channel (LAB-A) because the a-channel can highlight S pixels very well. After normalizing the a-channel to the range of 0 to 255, we used Otsu threshold [28] to set the H pixel to gray and the S pixel to white. Finally, a three-color pixel label for training was completed.

The WSR images contain long and thin leaves with regular margins and spots with complex distribution (Figure 1b). We tried to continue using the soybean bacterial blight scheme but found that the LAB-A method labeled many *S* pixels as *H* pixels. Instead of the LAB-A-highlighted diseased spots, we used the healthy leaves highlighted by the modified green–red vegetation index (MGRVI) [29].

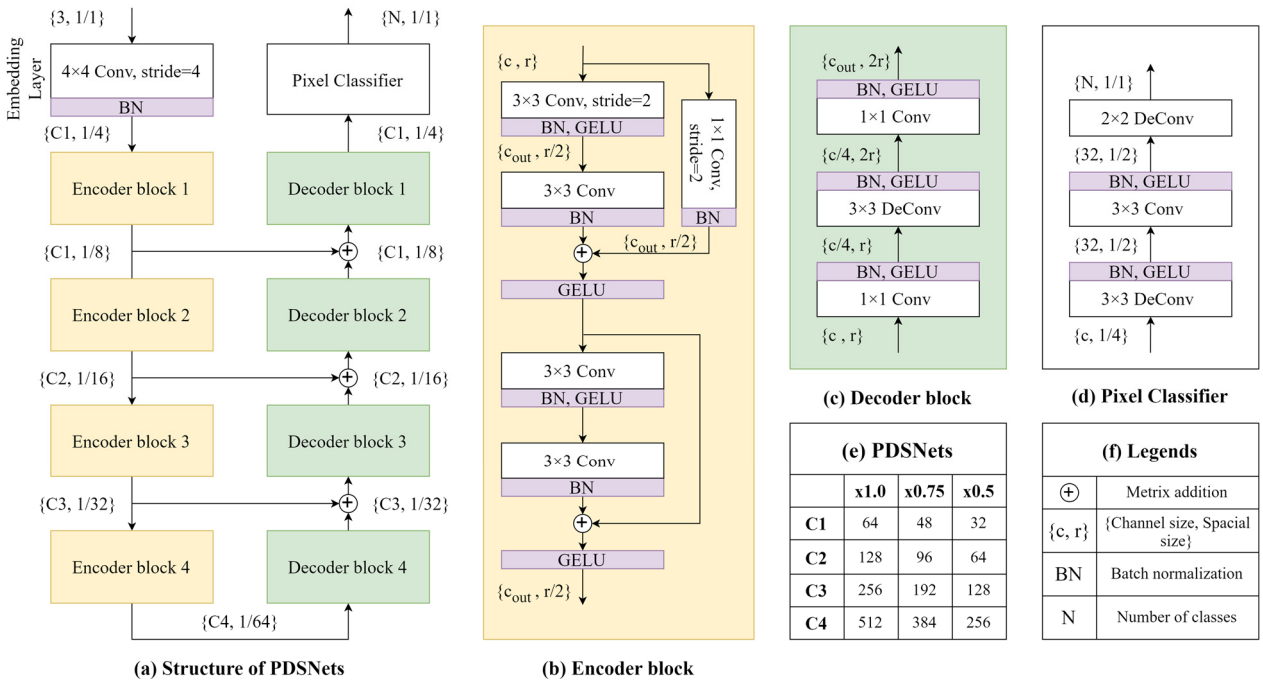
$$MGRVI = \frac{G^2 - R^2}{G^2 + R^2} \quad (2)$$

where *R* and *G* are the green and red channels of the RGB color space, respectively. Their value range is from 0 to 1. Before performing threshold segmentation, the MGRVI also needs to be normalized to the range of 0 to 255. In this way, we successfully labeled all the *S* pixels, but at the same time, the white edges and leaf stalks of wheat leaves in some images were incorrectly labeled as *S* pixels. Therefore, we used morphological opening operation to remove these errors.

The CAR images contain leaves and spots with regular margins, but reflections and shadows caused by direct sunlight render the above scheme ineffective (Figure 1c). So, we used LabelMe to manually annotate the edge polygon of a main leaf and the spots on it and then converted it into the required three-color pixel label.

2.3. PDSNets

Figure 2 presents the Plant Disease Segmentation Networks (PDSNets). They are based on LinkNet, using a new 4 × 4 embedding layer and the GELU activation function [30]. To represent feature size simply, we used the fraction between its height (width) and the height (width) of the input image. For example, a feature with a size of {64, 1/4} means that its channel size is 64 and its height (width) is 1/4 of the height (width) of the input image.



(e) PDSNets

	x1.0	x0.75	x0.5
C1	64	48	32
C2	128	96	64
C3	256	192	128
C4	512	384	256

(f) Legends

⊕	Metrix addition
{c, r}	{Channel size, Spacial size}
BN	Batch normalization
N	Number of classes

Figure 2. The structure of Plant Disease Segmentation Networks (PDSNets). The spatial size is defined as the fraction between the height (width) of the feature and the height (width) of the input image.

The encoder of PDSNets consists of a 4 × 4 embedding layer and 4 encoder units. It can extract image features step by step as the channel size of the features increases and the spatial size decreases. The 4 × 4 embedding layer, which can effectively increase the processing speed, is responsible for mapping the RGB image into the function domain defined by the network. Each encoder unit consists of two residual blocks from ResNet [8], with the first changing the size of the feature and the second keeping the feature size unchanged (Figure 2b). The size of the highest-level feature in the final output of the encoder is {512, 1/64}.

The decoder of PDSNets is made of 4 decoder units and a pixel-level classifier. Each decoder unit uses two 1 × 1 convolutional layers to reduce and increase the channel size of the features, making the 3 × 3 deconvolution operation [31] in the middle cheaper to compute (Figure 2c). Each decoder unit accepts not only the output features of the previous decoder but also the output features of the encoder of the same level, also known as skip connection. The pixel-level classifier uses two deconvolutions to restore the spatial size of the features to the image size (Figure 2d). In the network, due to the usage of batch normalization after each convolutional layer, we removed the bias from all convolutional and deconvolutional layers except the last deconvolutional layer of the pixel-level classifier.

As shown in Figure 2e, we scaled the channel size of PDSNetx1.0 to obtain its lightweight versions. The channel sizes of PDSNetx0.75 are 0.75 times the baseline, meaning its 4 encoders have channel sizes of 48, 96, 192, and 384, respectively. Similarly, the channel sizes of PDSNetx0.5 are 0.5 times the baseline, meaning its 4 encoders have channel sizes of 32, 64, 128, and 256, respectively. During the scaling process, the number of input and output channels of the intermediate 3 × 3 convolutional layer in the pixel-level classifier is fixed at 32.

2.4. Evaluation Indices

*F*1 and *IoU* were used to evaluate the pixel classification performance of the network. For pixels labeled as *i*, *F*1 can evaluate the accuracy of pixel classification, and *IoU* can

assess the similarity between true label i pixels and the predicted label i pixels. They are defined as follows:

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{5}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{6}$$

where *true positive* (*TP*) is the number of pixels where both the true label and the predicted label are i, *false positive* (*FP*) is the number of pixels where the predicted label is i but the true label is not i, and *false negative* (*FN*) is the number of pixels where the true label is i but the predicted label is not i. Therefore, *precision* represents the proportion of pixels with the predicted label i that were correctly predicted, and *recall* represents the proportion of pixels with the true label i that were correctly predicted. $F1$ is the harmonic mean of precision and recall.

The coefficient of determination (R^2) and root mean square error (RMSE) between the reference severity and the predicted severity were used to evaluate the network's performance in estimating severity. In addition, scatter plots and linear regression lines were drawn for auxiliary evaluation. FPS is used to evaluate the prediction speed of the network, indicating the number of 640 × 640 pixel images the network can predict per second.

$$FPS = \frac{1}{T} \tag{7}$$

where T is the average time taken for the network to make 300 predictions on the local i5-12400 CPU.

2.5. Experiment Settings

All experiments in this work were conducted on the same computer, which is equipped with Windows 11 (10.0.26100, Microsoft Corporation, Redmond, WA, USA), Pytorch 12.4, Intel i5-12400 CPU (Intel Corporation, Santa Clara, CA, USA), and NVIDIA GeForce 4060ti GPU (Nvidia Corporation, Santa Clara, CA, USA). The maximum number of epochs and the batch size for network training are 200 and 8, respectively. The learning rate (*LR*) is 0.0003, and it is reduced every epoch.

$$LR = \left(1 - \frac{epoch}{epoch_{max}}\right)^{0.9} \tag{8}$$

The optimizer is AdamW [32] with a default decay rate of 0.01. The weighted cross-entropy loss (WCE-Loss) was used to balance the naturally occurring pixel count imbalance between B, H and S. The WCE-Loss for a single pixel is defined as follows:

$$WCE\text{-}Loss = -\sum_{i=1}^{N} \omega_i \cdot p_i \cdot log\left(\frac{\exp(q_i)}{\sum_{j=1}^{N} \exp(q_j)}\right) \tag{9}$$

where p and q are the one-hot true label and the predicted logits for the pixel, respectively. N is the number of pixel classes. We empirically set the weight as B: $\omega_1 = 1$; H: $\omega_2 = 3$; and S: $\omega_3 = 6$.

For each disease, we randomly selected 60% of the data for training and the remaining 40% for testing. During the training process, we used random image transformations [33]

to make the training data different for each epoch, which increased the diversity of the training data. These image transformation methods included random rotations within the range of -25° to 25° and random flips up, down, left, and right. The probability of applying these image transformation methods to each input image was 0.5. The training data were shuffled and combined into new mini-batches at the start of each epoch. In addition, when training the U-Net, we set the batch size to 4 to prevent running out of GPU memory. For the backbone of the DeepLabv3+, we used the dilated ResNet-18 with the dilation method described in [19].

3. Results

3.1. Comparison of Networks

As shown in Table 2, we compared the segmentation results of the PDSNets with the common U-Net, DeepLabv3+ with the ResNet-18 backbone, and LinkNet under the same experimental conditions. The *F*1 scores and *IoU* values in the table indicate the cross-disease segmentation accuracy of these networks for *B*, *H*, and *S* classes. Params represent the numbers of learnable parameters, and FPS_{CPU} indicates their prediction speed on the local CPU.

Table 2. The comparison of networks under field scenarios. *B*, *H*, and *S* represent the background, healthy leaf, and disease spots, respectively. FPS_{CPU} indicates how many 640 × 640 pixel images can be processed per second on the local central processing unit (CPU). The F1 score and IoU values are shown as the means across diseases.

Networks	*F*1 (%)			*IoU* (%)			Params (M)	FPS_{CPU} (F/s)
	B	*H*	*S*	*B*	*H*	*S*		
U-Net	98.24	91.56	89.14	96.59	85.41	80.83	31.04	0.549
DeepLabV3+(ResNet-18)	98.66	93.05	87.15	97.40	87.53	77.52	16.60	2.673
LinkNet	98.80	93.05	86.19	97.66	87.54	76.07	11.54	12.86
PDSNetx1.0	98.55	93.30	86.38	97.19	87.71	76.34	11.53	19.88
PDSNetx0.75	98.46	92.74	85.37	97.03	86.91	74.88	6.497	25.78
PDSNetx0.5	98.49	92.52	84.87	97.07	86.37	74.11	2.898	34.18

To evaluate the segmentation performance of these networks, we averaged the *F*1 score and *IoU* values across pixel classes in the table. U-Net achieved the highest *F*1 and *IoU* values at 92.98% and 87.61%, respectively, but its prediction speed is only half an image per second. Compared to U-Net, DeepLabv3+(ResNet-18) has fewer parameters, but its prediction speed is still very low. As the benchmark for our work, the prediction speed of LinkNet is 23.4 times that of U-Net, with only a 0.3% *F*1 and 0.52% *IoU* cost. The PDSNetx1.0 we proposed has a prediction speed 1.55 times that of LinkNet with almost no cost to accuracy. The prediction speed of the lightest PDSNetx0.5 is 2.66 times that of LinkNet with costs of 0.72% for *F*1 and 1.24% for *IoU*.

To evaluate the segmentation performance of CNNs on the pixel classes *B*, *H*, and *S*, we averaged the *F*1 and *IoU* values across networks in the table. The *F*1 and *IoU* values for class *B* are 98.53% and 97.16%, respectively; for class *H*, they are 92.7% and 86.91%, respectively; and for class *S*, they are 86.52% and 76.63%, respectively.

In summary, our PDSNets achieved the best balance between prediction speed and accuracy. CNNs have the highest segmentation ability for the background, followed by healthy leaf, and the lowest for disease spots [34].

Figure 3 visually demonstrates the prediction results of the networks, showing their excellent ability to extract target diseased leaves from field background images. However, in complex scenarios with multiple leaves, reflections, and shadows, they exhibited lower

segmentation accuracy (Figure 4). Since most images in our experiments are single subject leaves and avoid direct sunlight, we cannot yet evaluate the networks' true ability to handle such complex scenarios.

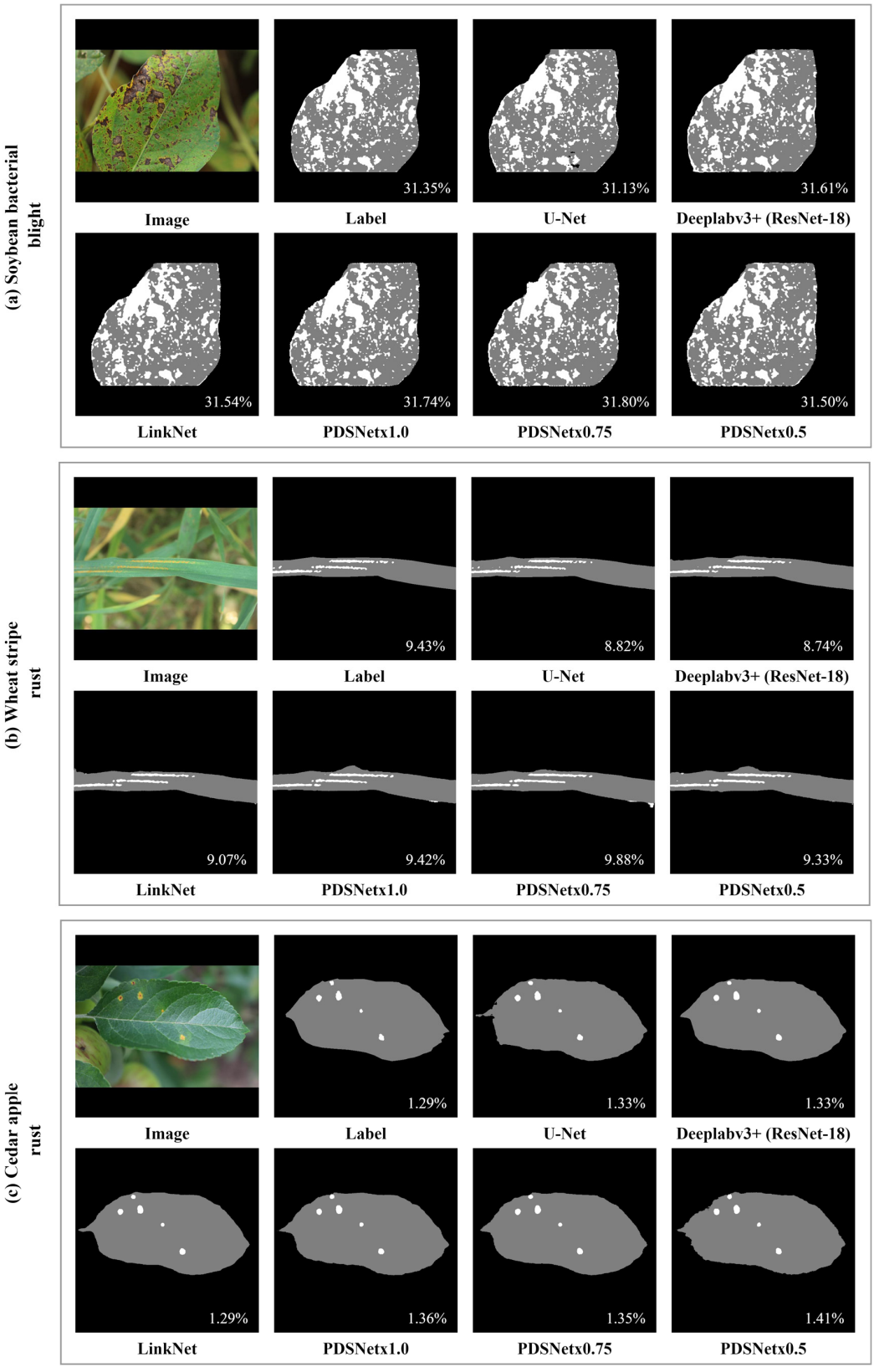


Figure 3. Examples of segmentation in conventional field scenarios. The percentage is the severity defined as white × 100/(white + gray).

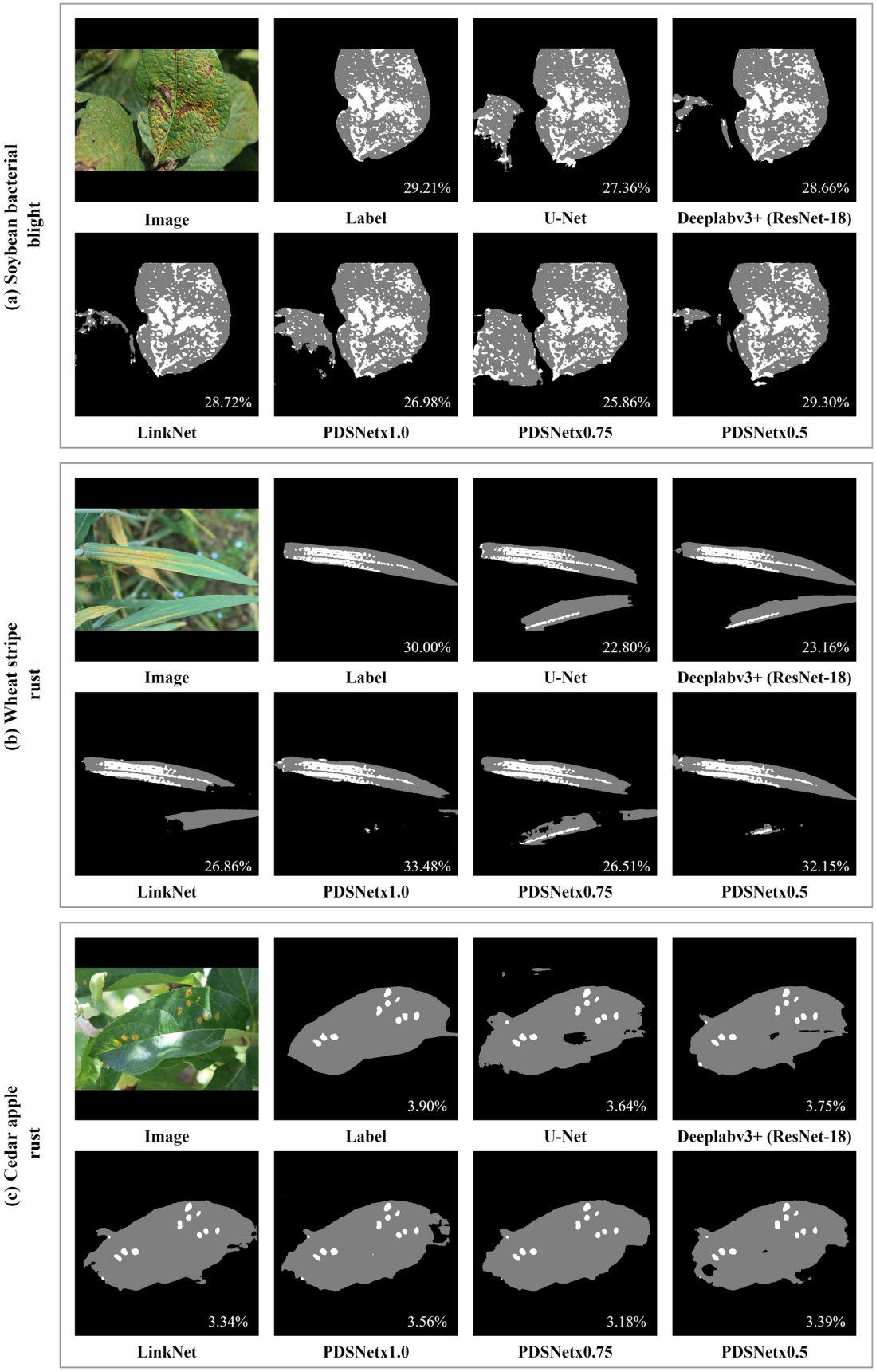


Figure 4. Examples of segmentation of complex field scenarios with multiple leaves, reflections, and shadows. The percentage is the severity defined as white × 100/(white + gray).

3.2. Ablation Experiment of PDSNetx1.0

As shown in Table 3, we conducted ablation experiments under field scenarios to verify the effects of the 4 × 4 embedding layer and the GELU activation function. The *F*1 and *IoU* in the table indicate the cross-disease segmentation accuracy of these networks for *B*, *H*, and *S*. FPS_{CPU} indicates their prediction speed on the local CPU. We took the average of *F*1 and *IoU* values across pixel classes. After applying the 4 × 4 embedding layer, the network's prediction speed was 1.58 times that of LinkNet with only a 0.17% cost for *F*1 and 0.38% cost for *IoU*. After simultaneously applying the 4 × 4 embedding layer and the GELU activation function, the model's prediction speed slightly decreased but was still 1.55 times that of LinkNet, while the model's segmentation accuracy slightly improved.

Table 3. The ablation results of PDSNetx1.0 on field-acquired images. *B*, *H*, and *S* represent the background, healthy leaf, and disease spots, respectively. FPS_{CPU} indicates how many 640 × 640 pixel images can be processed per second on the local central processing unit (CPU). The F1 score and IoU values are shown as means across diseases. "−" indicates below the baseline and "+" indicates above the baseline.

Networks	*F*1 (%)			*IoU* (%)			FPS_{CPU} (F/s)
	B	*H*	*S*	*B*	*H*	*S*	
LinkNet + ReLU (Baseline)	98.80	93.05	86.19	97.66	87.54	76.07	12.86
LinkNet + 4 × 4 embedding + ReLU	98.56 (−)	92.77 (−)	86.20 (+)	97.20 (−)	86.94 (−)	76.00 (−)	20.37 (+)
LinkNet + 4 × 4 embedding + GELU (PDSNetx1.0)	98.55 (−)	93.30 (+)	86.38 (+)	97.19 (−)	87.71 (+)	76.34 (+)	19.88 (+)

In summary, the 4 × 4 embedding layer can effectively improve the model's prediction speed, while the GELU activation function effectively compensates for the segmentation accuracy loss brought by the 4 × 4 embedding layer.

3.3. CE-Loss vs. WCE-Loss of PDSNetx1.0

As shown in Figure 5, we recorded the training information of PDSNetx1.0 using cross-entropy loss (CE-Loss) and weighted cross-entropy loss (WCE-Loss), including the training loss and test *F*1 score for each disease in each training epoch. We also calculated the proportion of *B*, *H*, and *S* pixel classes in each disease training set. The results for SBB are *B*:69.55%, *H*:22.77%, and *S*:7.68%; for WSR, they are *B*:90.65%, *H*:7.30%, and *S*:2.05%; and for CAR, they are *B*:75.90%, *H*:23.66%, and *S*:0.44%. The improvement in WCE-Loss on PDSNetx1.0 in SBB and WSR is not significant, with the cross-class average *F*1 score increasing by 0.66% and 0.92%, respectively, and the *F*1 score of *S* pixels increasing by 2.03% and 1.80%. However, the improvement in WCE-Loss on CAR is significant, especially in the case of disease spots. The cross-pixel class average *F*1 score of CAR improved by 2.74%, with the *F*1 score of *S* pixels improving by 7.52%.

In summary, a very small number of samples will significantly increase the difficulty of the network's learning. WCE-Loss can effectively enhance PDSNetx1.0's classification performance of *S* pixels in disease images, particularly for diseases like CAR with spot-like distributions that result in too few *S* pixels, where the improvement is especially noticeable.

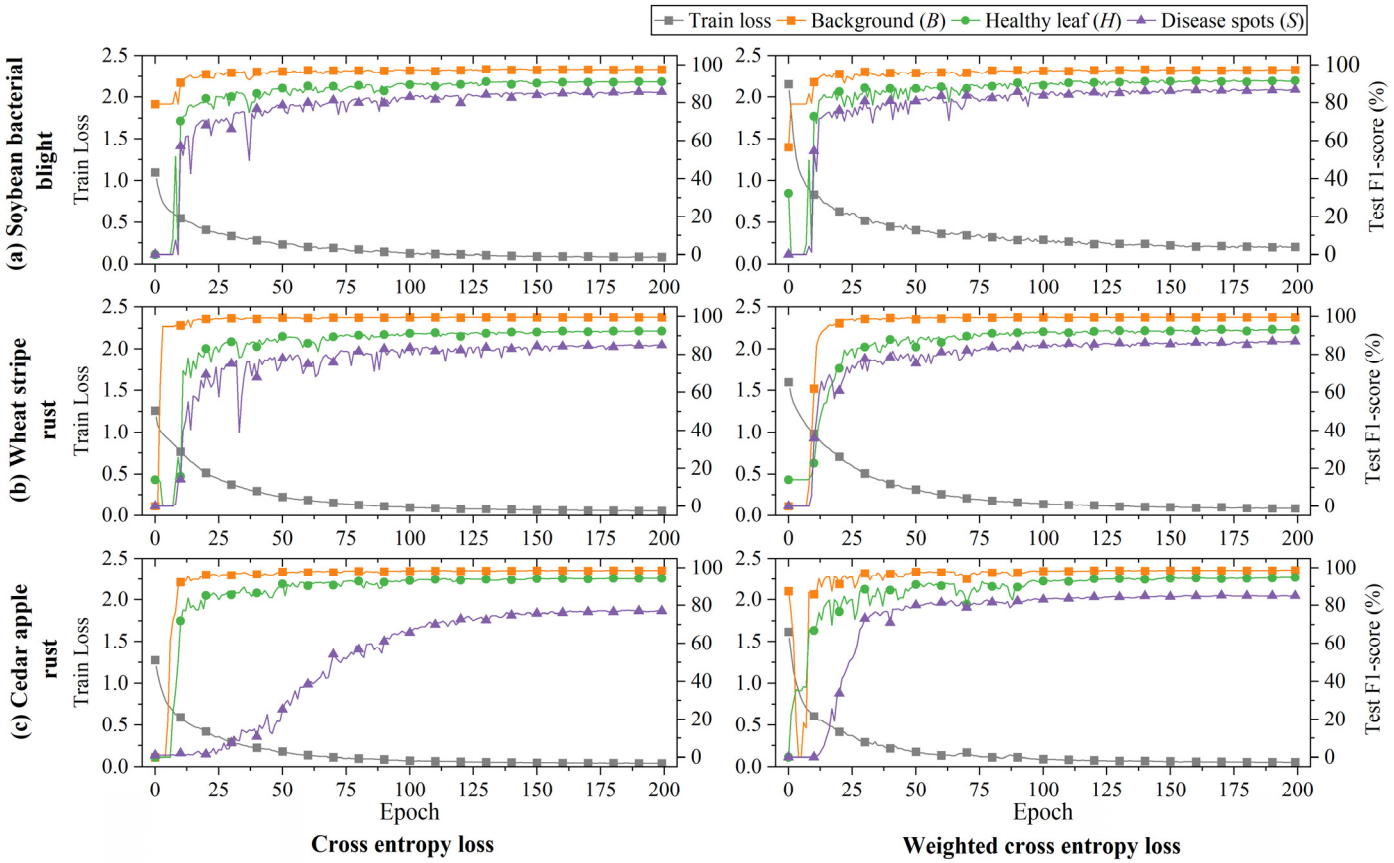


Figure 5. Comparison of PDSNetx1.0 using cross-entropy loss and weighted cross-entropy loss.

3.4. Generalization Test on Field, Laboratory, and Mixed Images

As shown in Figure 6, we used training sets of field, laboratory, and mixed images to train U-Net, DeepLabv3+(ResNet-18), LinkNet, and PDSNetx1.0, respectively, and then tested them on the test sets of field images and laboratory images. The mixed images are half field images and half laboratory images. Such an experimental setup can test the model's generalization ability under the field and laboratory scenarios.

To conduct a quantitative evaluation, we took the cross-network average of the *F*1 for each disease. For SBB, the networks trained with field images performed similarly in lab image tests, but the networks trained with lab images decreased by 32.65% in field image tests. For WSR, the networks trained with field images decreased by 26.76% in lab image tests, and the networks trained with lab images decreased by 60.97% in field image tests. For CAR, the networks trained with field images decreased by 29.36% in lab image tests, and the networks trained with lab images decreased by 40.87% in field image tests. The networks trained with mixed images performed well in the tests of two background images for each disease.

In summary, CNNs cannot accurately predict data that are not present in the training set, even though the lab images in this work are merely simulated results of field images with the background removed. It can be reasonably inferred that if images are collected from real laboratory scenarios with greater differences, the CNNs will be even less capable of carrying out cross-background prediction.

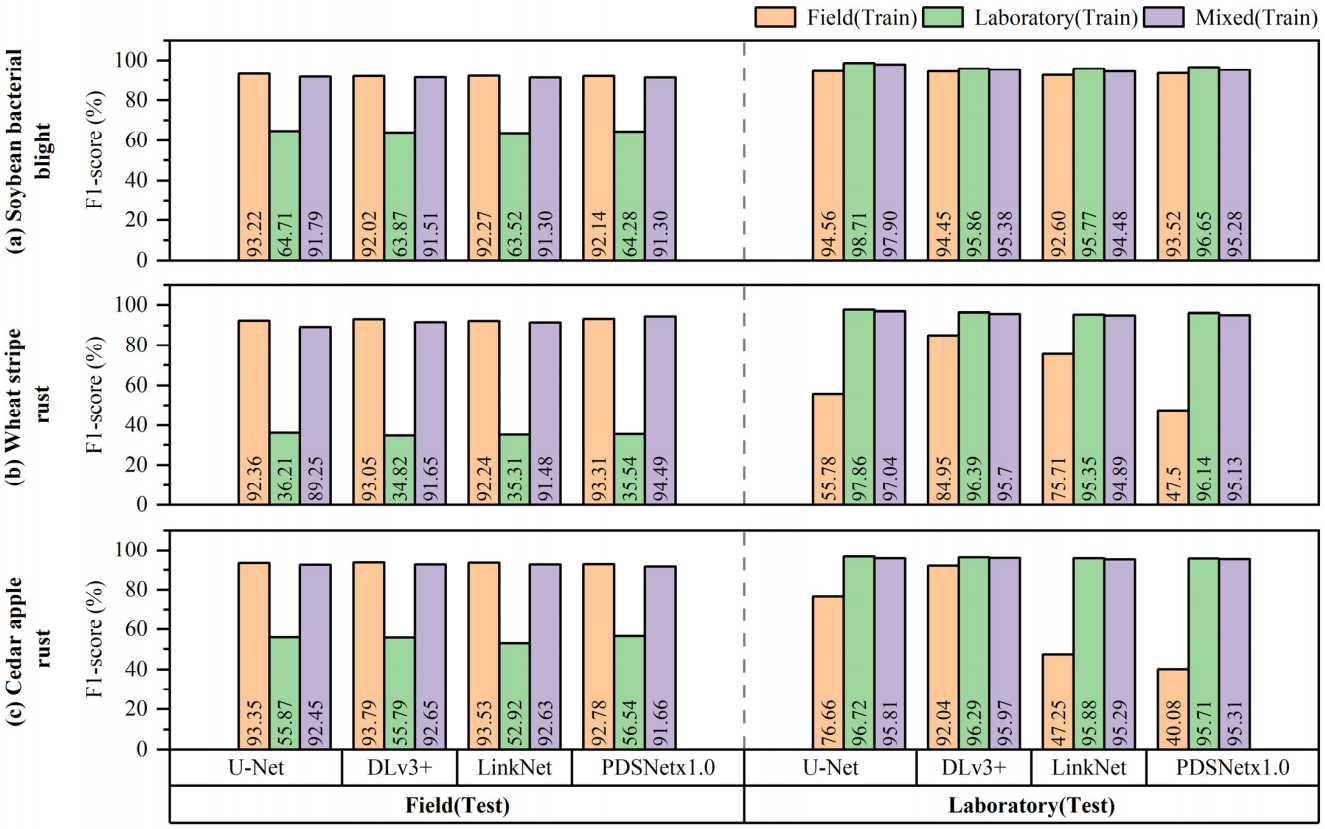


Figure 6. Test results of the networks' generalization ability under field and laboratory scenarios. DLv3+ is the abbreviation for DeepLabv3+(ResNet-18).

3.5. Severity Estimation

To evaluate the severity estimation capability of the networks under field scenarios, we calculated the reference severity of the labels and the predicted severity of the networks. Severity refers to the proportion of a leaf occupied by disease spots, calculated using Equation (1).

As shown in Figure 7, the average R^2 and RMSE values of SBB are 0.888 and 2.942, respectively. DeepLabV3+ achieved the best estimation results, but the differences among these networks were not significant. According to the regression lines, most networks overestimated the severity. As shown in Figure 8, the average R^2 and RMSE values of WSR are 0.966 and 2.711, respectively. Like SBB, DeepLabV3+ achieved the highest estimation performance, but the differences among the networks were not significant. Most models overestimated severity, while U-Net underestimated it. As shown in Figure 9, the average R^2 and RMSE values of CAR are 0.921 and 0.331, respectively. LinkNet achieved the best estimation results with little difference among the networks. Most networks underestimated the severity.

In summary, without considering the number of parameters and prediction speed, DeepLabv3+(ResNet-18) demonstrated the best estimation performance. The proposed PDSNetx0.5, with its parameter count and prediction speed being 0.175 times and 12.79 times that of DeepLabv3+, respectively, only saw a reduction in R^2 for the three diseases by 0.069 for SBB, 0.026 for WSR, and 0.029 for CAR. Therefore, PDSNetx0.5 achieved a good balance between prediction accuracy and speed in severity estimation.

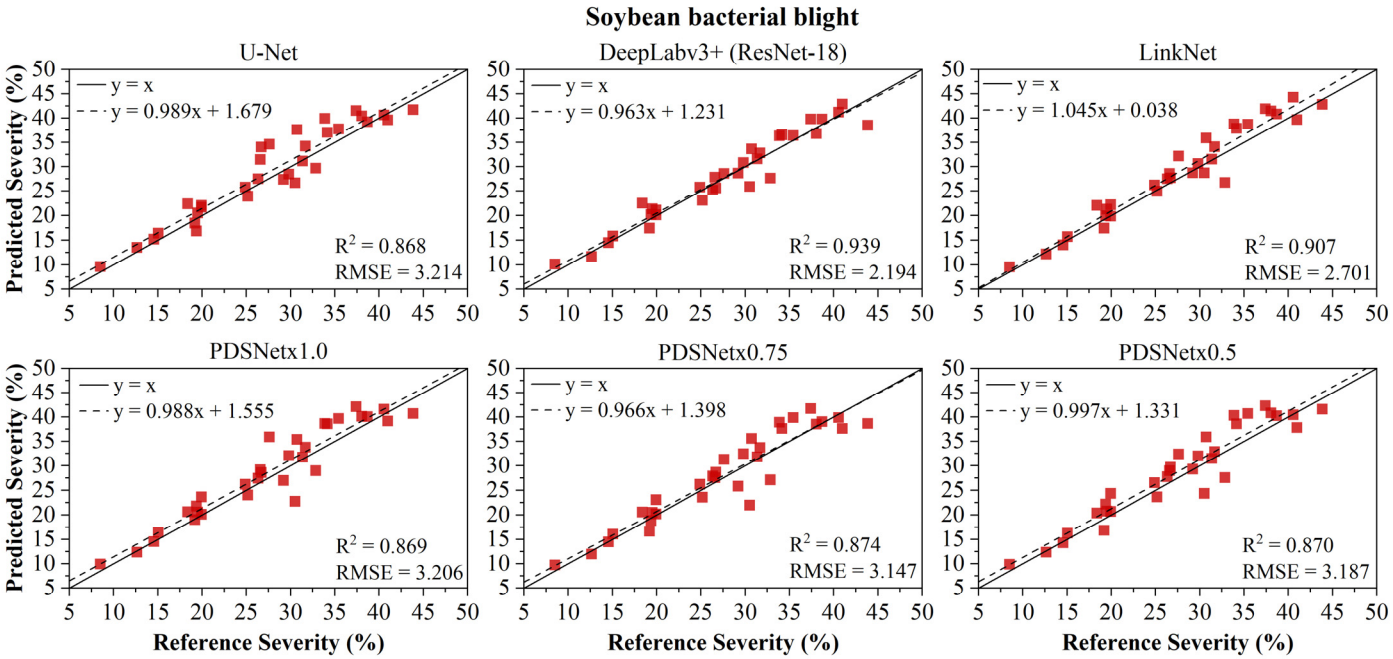


Figure 7. The severity estimation results of soybean bacterial blight. The dashed line is the linear regression line, and the solid line is the best fit line.

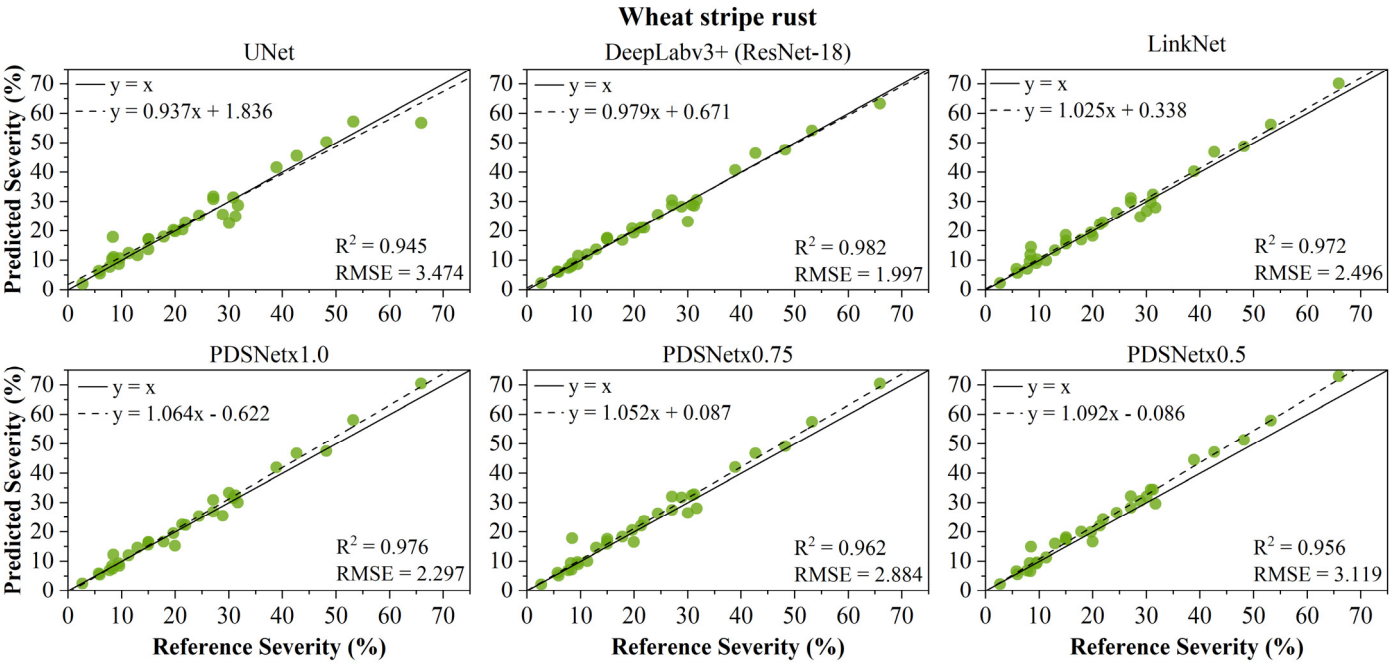


Figure 8. The severity estimation results of wheat stripe rust. The dashed line is the linear regression line, and the solid line is the best fit line.

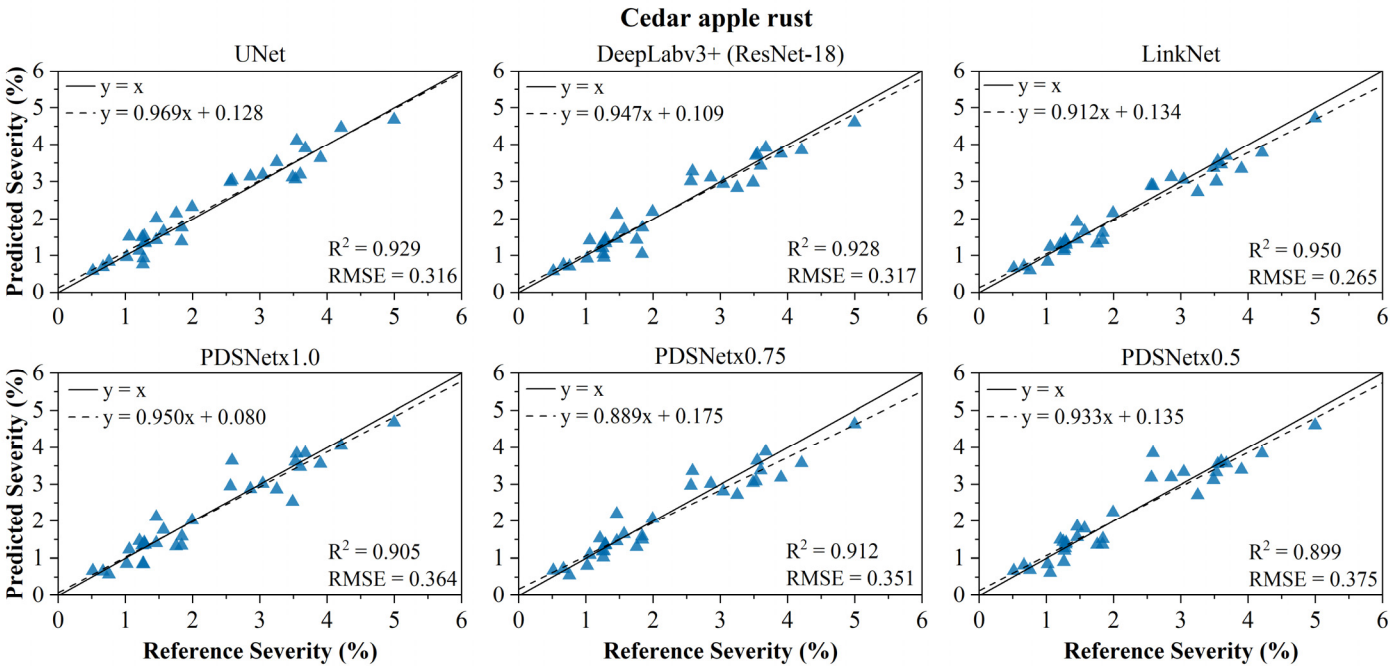


Figure 9. The severity estimation results of cedar apple rust. The dashed line is the linear regression line, and the solid line is the best fit line.

4. Discussion

4.1. Lightweight Networks

In this study, we introduced the PDSNets family, built upon the LinkNet architecture, which employs ResNet-18 as the encoder and incorporates a 4 × 4 embedding layer to enhance the prediction speed by 50%. As a flagship model of the PDSNets family, PDSNetx0.5 demonstrates exceptional performance in segmenting multiple diseases, achieving an overall *F*1 score of 91.96% and an Intersection over Union (*IoU*) of 85.85%, alongside a coefficient of determination (R^2) of 0.908 for disease severity evaluation. With a compact parameter count of only 2.9 M, PDSNetx0.5 represents just 25.1% and 9.33% of the parameter sizes of LinkNet and U-Net, respectively. In contrast to existing lightweight network research [21,22,24], PDSNetx0.5 has been rigorously tested on diverse disease images, showcasing superior generalization capabilities. Furthermore, the implementation of the weighted cross-entropy loss function has been shown to significantly improve the network's ability to capture lesion characteristics [18].

4.2. Limitations

The primary limitation of PDSNetx0.5 stems from the reduction in network parameters, resulting in a decline in prediction accuracy. To address this, it is essential to incorporate advanced deep learning techniques, such as attention mechanisms [22] and knowledge distillation [35], which can significantly enhance the accuracy of lightweight networks. Additionally, the limited sample size, particularly the absence of data from multi-leaf and complex lighting scenarios, constrains the network's generalization capabilities. Moreover, the model has not been deployed on mobile platforms for real-world testing, further limiting its practical applicability.

The experimental findings of this study demonstrate that networks trained in laboratory scenarios cannot be directly utilized for field scenario predictions, and the opposite is also true. This observation holds across multiple mainstream architectures, including U-Net, DeepLabV3+, and LinkNet. These results underscore the critical importance of developing field scenario datasets to advance research in field applications.

4.3. Future Work

In future studies, we will prioritize the following research directions: (1) integrating advanced deep learning techniques, including attention mechanisms and knowledge distillation, to further enhance the accuracy of PDSNetx0.5; (2) expanding the scale and diversity of the field scenario dataset by increasing the sample size and incorporating additional disease types, thereby improving the network's generalization capabilities; and (3) deploying PDSNetx0.5 on mobile platforms for real-world testing to evaluate its practical performance. Overall, the end-to-end design enables farmers to utilize the tool effectively without specialized technical expertise. Furthermore, the automated disease severity assessment approach not only significantly boosts the efficiency of field disease surveys conducted by farmers and experts but also offers robust support for optimizing agricultural production and disease management strategies.

5. Conclusions

In conclusion, this study introduced PDSNets, a family of lightweight semantic segmentation networks, with PDSNetx0.5 achieving remarkable performance in disease segmentation tasks. With only 2.9 M parameters, PDSNetx0.5 surpasses mainstream networks like LinkNet and U-Net in efficiency. The experimental findings reveal that networks trained in laboratory settings cannot be directly utilized in field scenarios, and the opposite is also true, underscoring the critical need for high-quality field datasets.

Author Contributions: Conceptualization C.Z. and C.L.; methodology, C.Z.; software, C.Z.; validation, Xin Wang.; investigation, C.Z., X.W. (Xin Wang), X.W. (Xifang Wu) and Y.D.; resources, C.Z. and C.L.; data curation, C.Z.; writing—original draft preparation, C.Z.; writing—review and editing, C.L.; visualization, C.Z. and Xin Wang; supervision, C.L., H.X. and Y.J.; funding acquisition, C.L., T.C. and H.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Fundamental Research Funds for the Universities of Henan Province, grant number 242300420221; the Fundamental Research Funds for the Universities of Henan Province, grant number 242300421365; and the National Major Scientific Research Achievement Cultivation Fund, grant number NSFRF240101.

Institutional Review Board Statement: Not applicable.

Data Availability Statement: The original data presented in this study are openly available in Kaggle at https://doi.org/10.34740/kaggle/ds/6620728, accessed on 9 March 2025.

Acknowledgments: The authors are thankful for the support provided by the Jiaozuo Academy of Agriculture and Forestry Sciences (http://www.jznlkxy.com/, accessed on 9 March 2025).

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

LAB-A	A-channel of LAB color space
MGRVI	Modified green–red vegetation index
CNN(s)	Conventional neural network(s)
SBB	Soybean bacterial blight
WSR	Wheat stripe rust
CAR	Cedar apple rust
B	Background
H	Healthy leaf
S	Disease spots

CPU	Central processing unit
GPU	Graph processing unit
FPS	Frame per second
GELU	Gaussian error linear unit
CE-Loss	Cross-entropy loss
TP	True positive
FP	False positive
FN	False negative
IoU	Interaction over union
LR	Learning rate

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